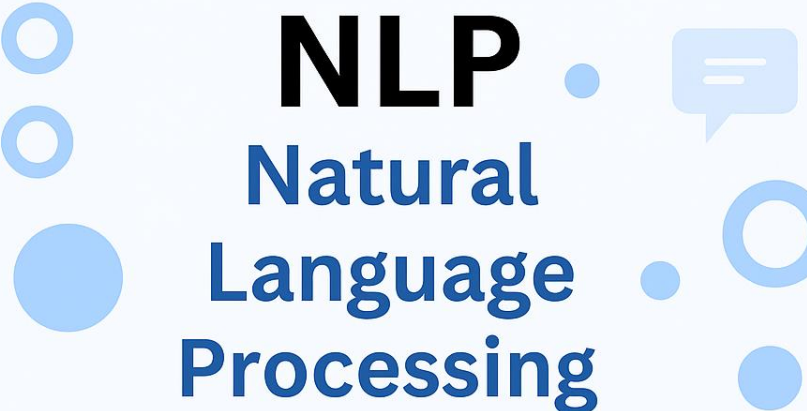
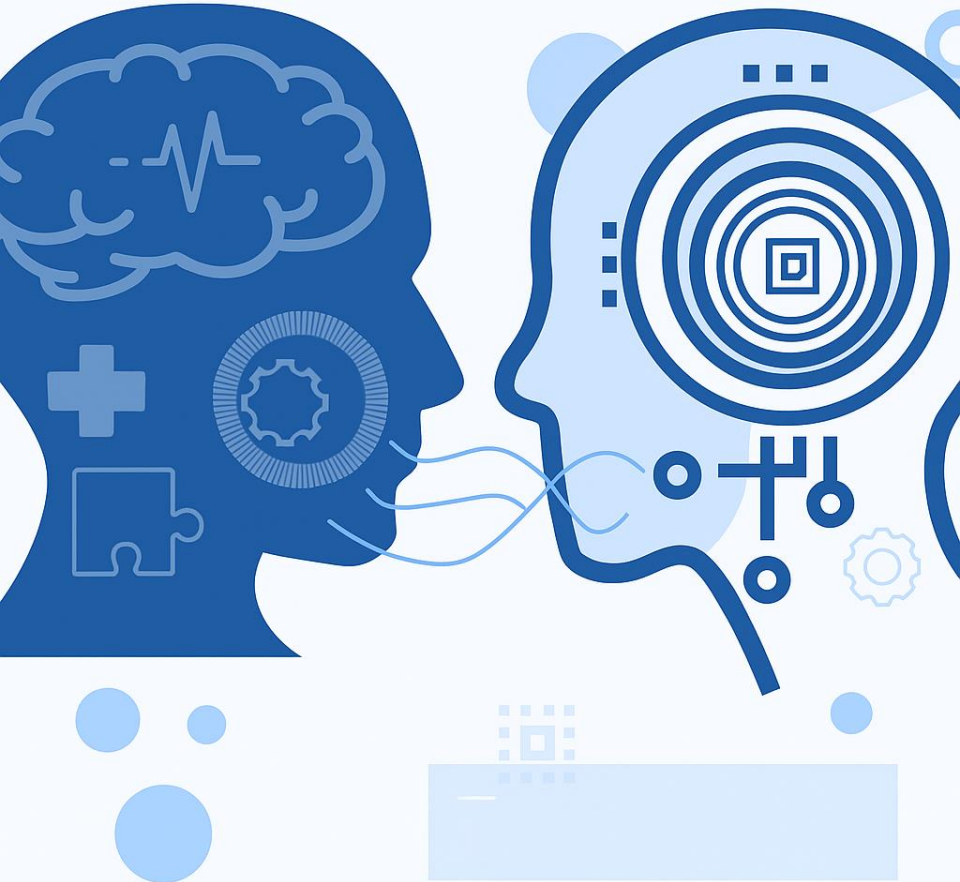




NLP

Natural
Language
Processing



NATURAL LANGUAGE PROCESSING (NLP)

PMDS606L

MODULE 4

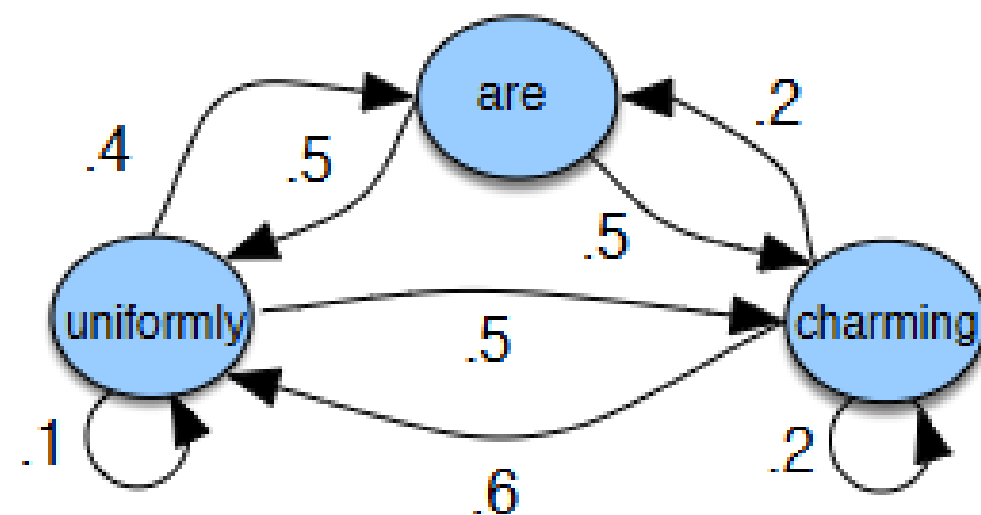
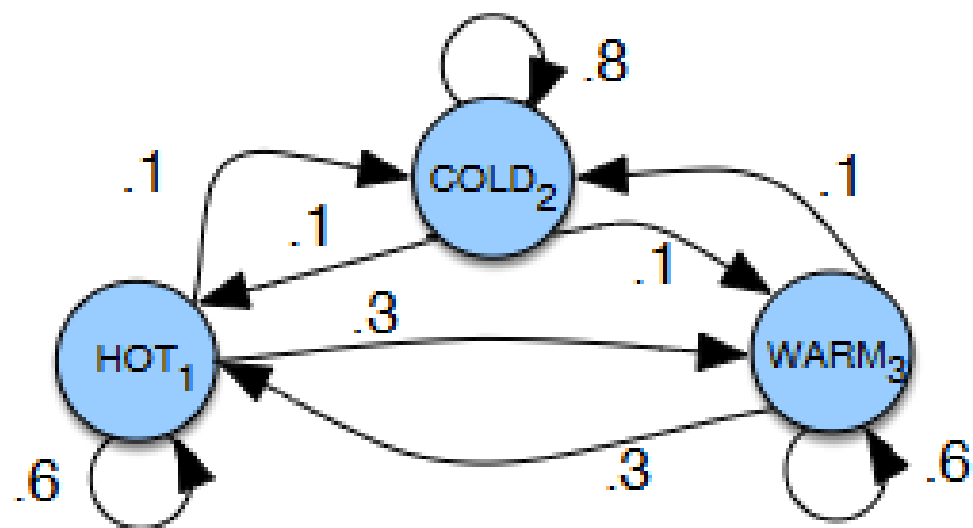
LECTURE 3

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$$Q = q_1 q_2 \dots q_N$$

a set of N states

$$A = a_{11} a_{12} \dots a_{n1} \dots a_{nn}$$

a **transition probability matrix** A , each a_{ij} representing the probability of moving from state i to state j , s.t. $\sum_{j=1}^n a_{ij} = 1 \quad \forall i$

$$\pi = \pi_1, \pi_2, \dots, \pi_N$$

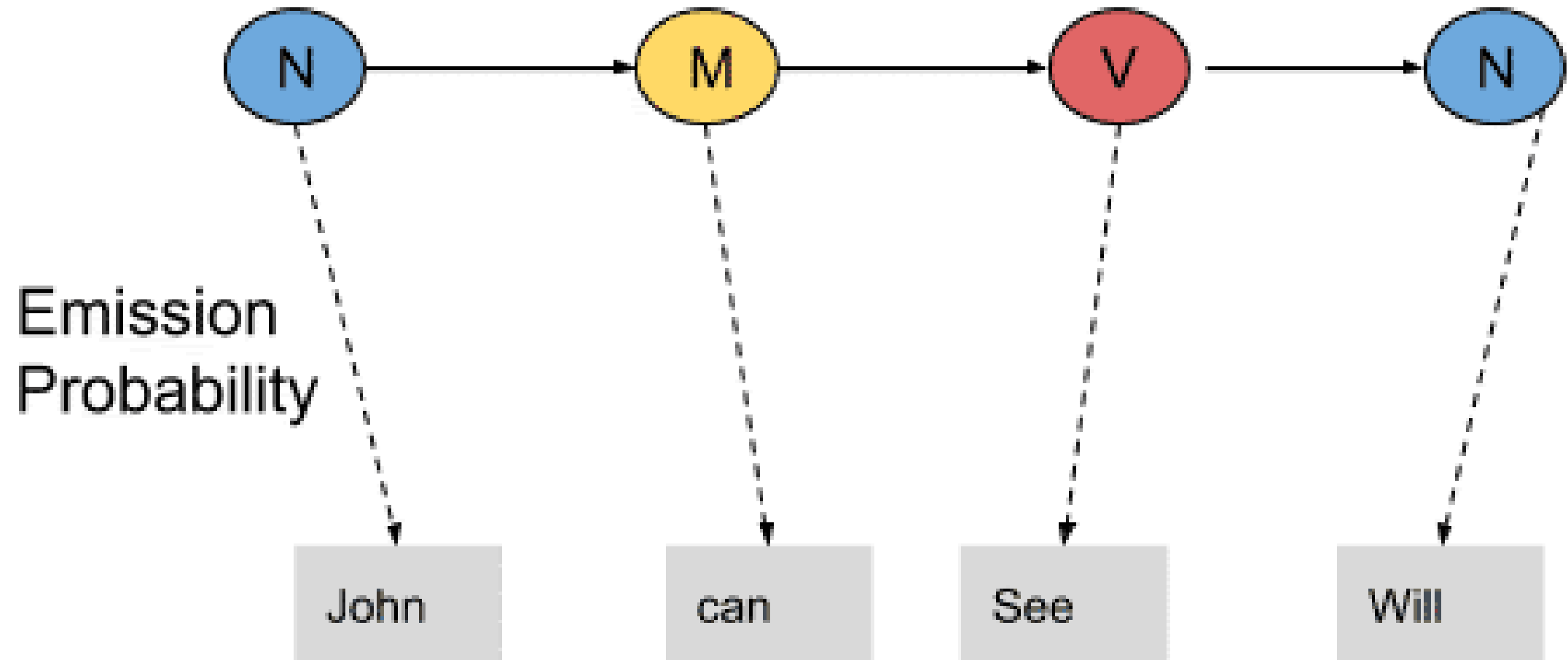
an **initial probability distribution** over states. π_i is the probability that the Markov chain will start in state i . Some states j may have $\pi_j = 0$, meaning that they cannot be initial states. Also, $\sum_{i=1}^n \pi_i = 1$

HMM in POS Tagging

We don't normally observe part-of-speech tags in a text. Rather, we see words, and must infer the tags from the word sequence. We call the tags hidden because they are not observed.

$Q = q_1 q_2 \dots q_N$	a set of N states
$A = a_{11} \dots a_{ij} \dots a_{NN}$	a transition probability matrix A , each a_{ij} representing the probability of moving from state i to state j , s.t. $\sum_{j=1}^N a_{ij} = 1 \quad \forall i$
$O = o_1 o_2 \dots o_T$	a sequence of T observations , each one drawn from a vocabulary $V = v_1, v_2, \dots, v_V$
$B = b_i(o_t)$	a sequence of observation likelihoods , also called emission probabilities , each expressing the probability of an observation o_t being generated from a state q_i
$\pi = \pi_1, \pi_2, \dots, \pi_N$	an initial probability distribution over states. π_i is the probability that the Markov chain will start in state i . Some states j may have $\pi_j = 0$, meaning that they cannot be initial states. Also, $\sum_{i=1}^n \pi_i = 1$

Transition Probability



N N M V N

Mary Jane can See Will

N M V N

Spot will see Mary

M N V N

Will Jane spot Mary ?

N M V N

Mary will pat Spot

Words	Noun	Model	Verb
Mary	4	0	0
Jane	2	0	0
Will	1	3	0
Spot	2	0	1
Can	0	1	0
See	0	0	2
pat	0	0	1

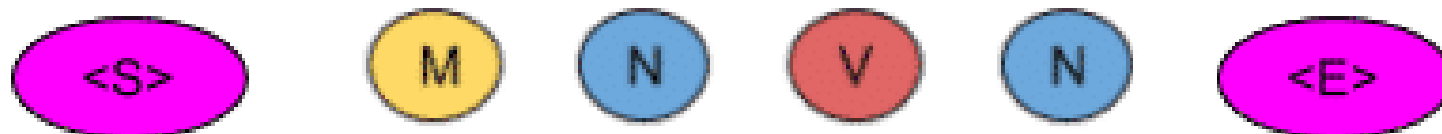
Words	Noun	Model	Verb
Mary	4/9	0	0
Jane	2/9	0	0
Will	1/9	3/4	0
Spot	2/9	0	1/4
Can	0	1/4	0
See	0	0	2/4
pat	0	0	1



Mary Jane can See Will



Spot will see Mary

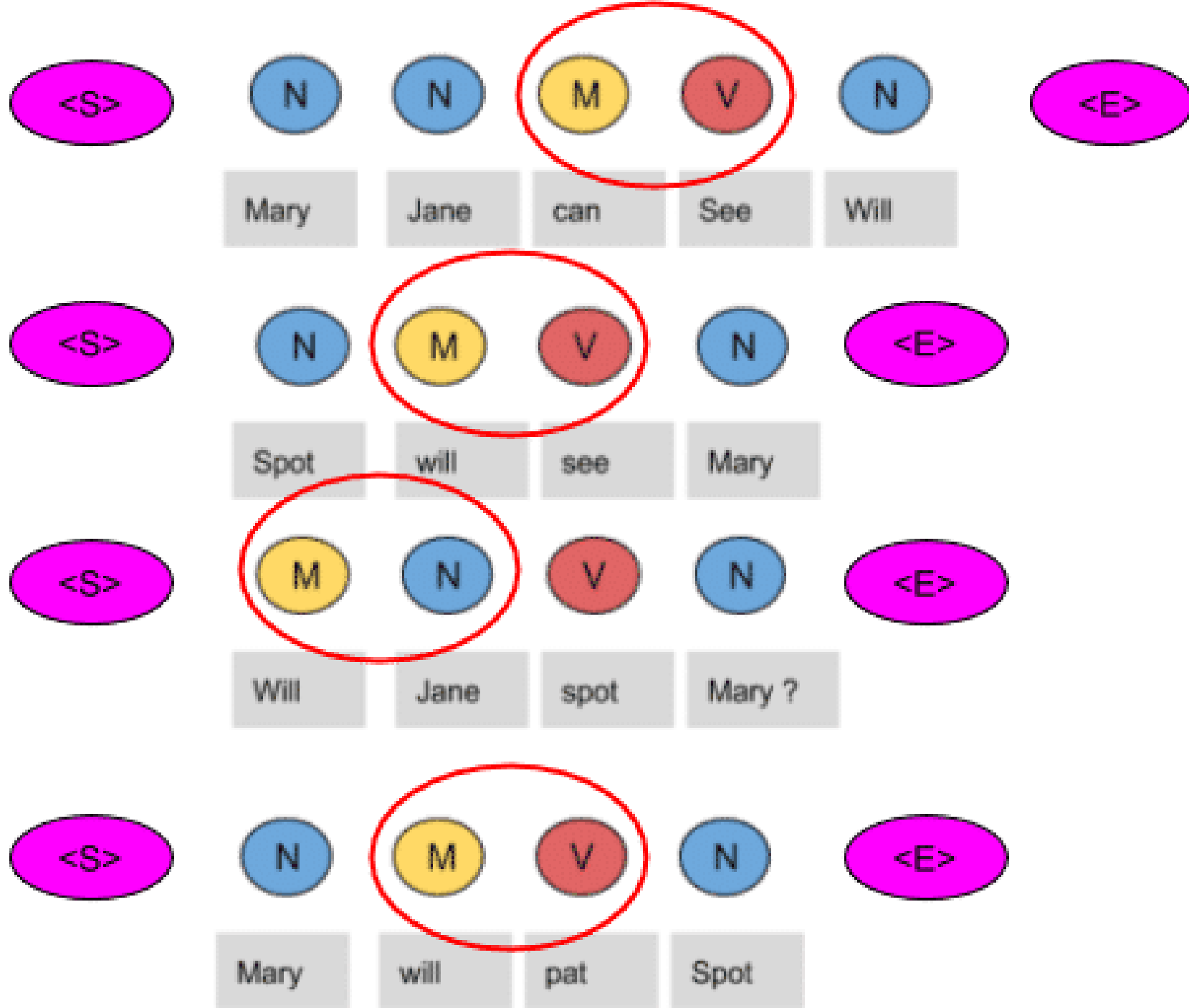


Will Jane spot Mary ?

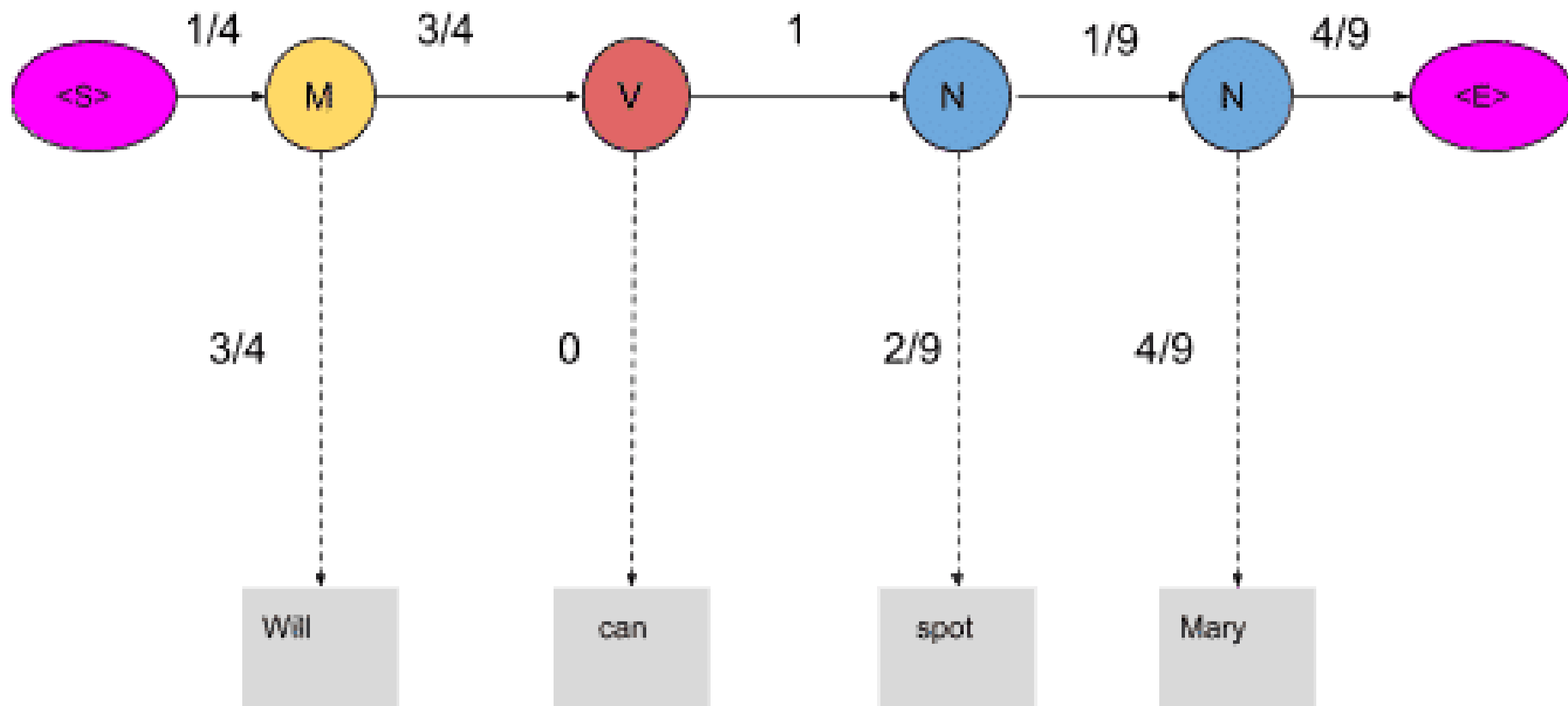


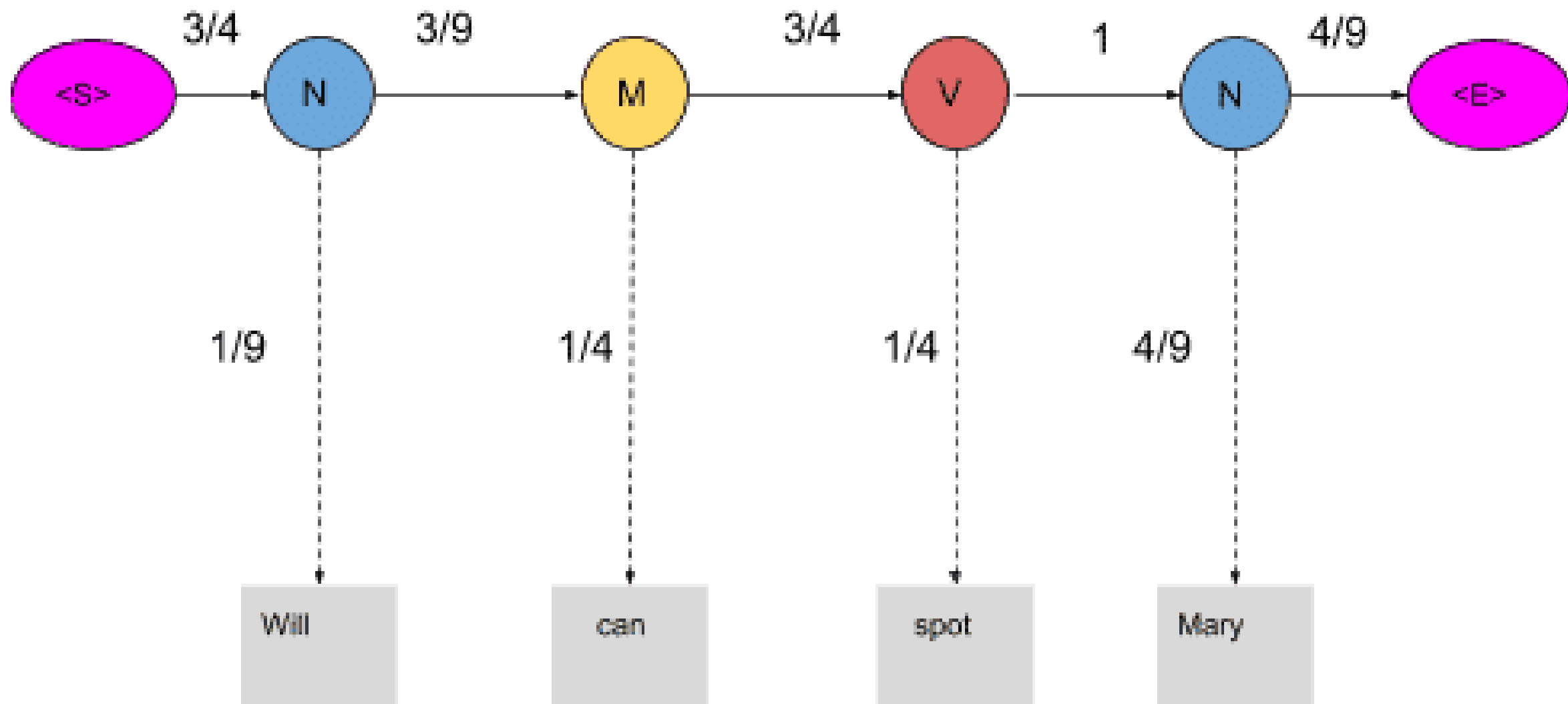
Mary will pat Spot

	N	M	V	<E>
<S>	3	1	0	0
N	1	3	1	4
M	1	0	3	0
V	4	0	0	0



	N	M	V	<E>
<S>	3/4	1/4	0	0
N	1/9	3/9	1/9	4/9
M	1/4	0	3/4	0
V	4/4	0	0	0

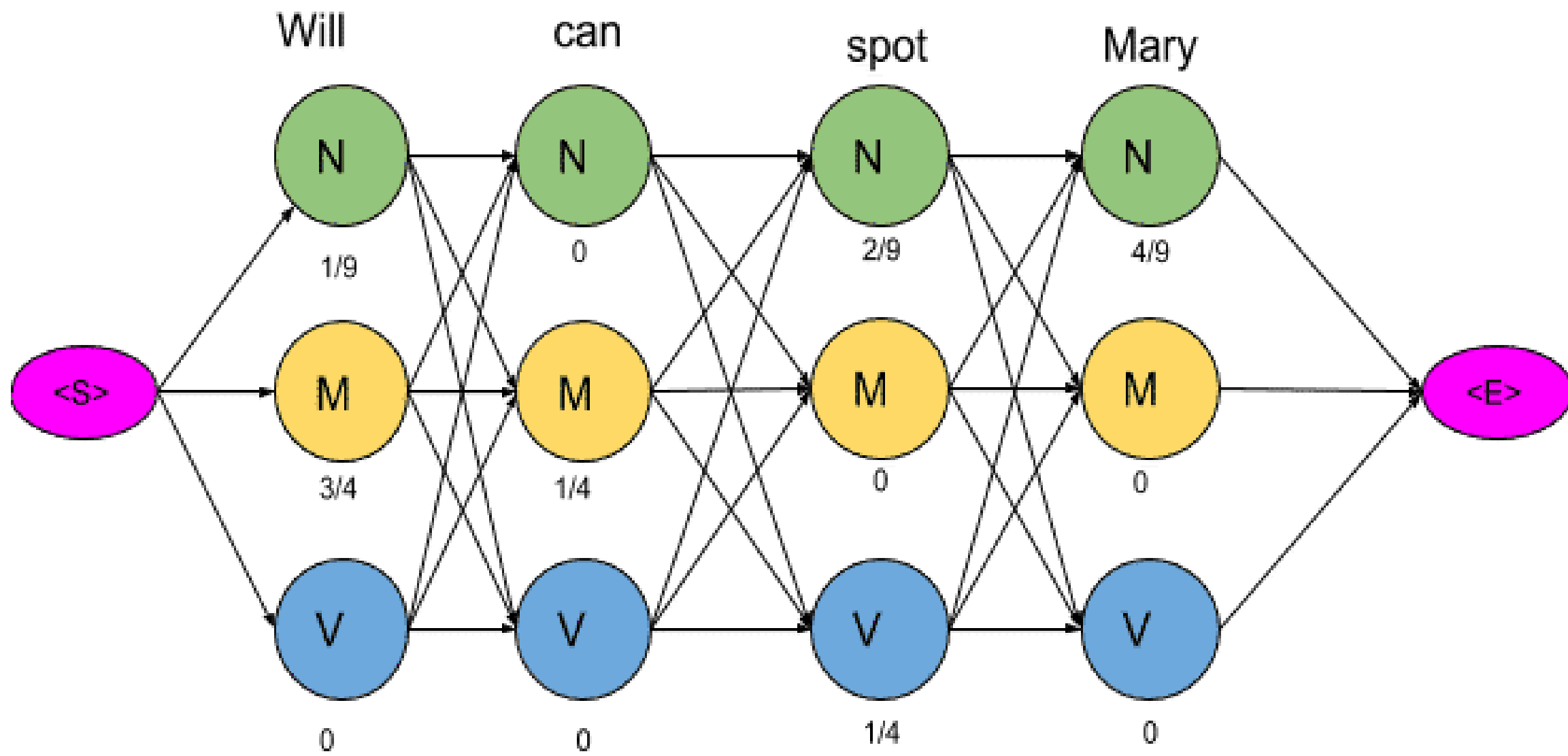


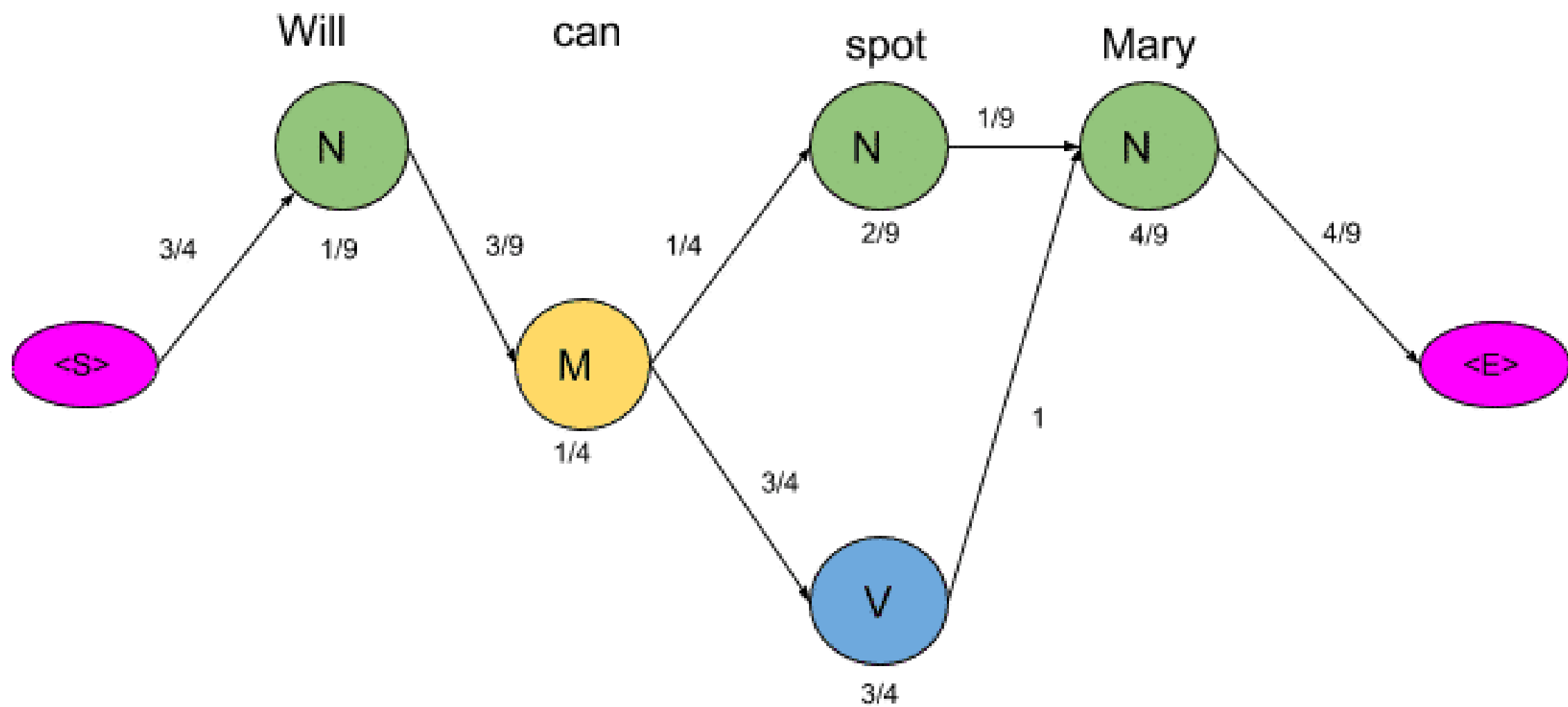


**AI is going to
take over the world**



**Is will a
noun or a verb ?**





	NNP	MD	VB	JJ	NN	RB	DT
$\langle s \rangle$	0.2767	0.0006	0.0031	0.0453	0.0449	0.0510	0.2026
NNP	0.3777	0.0110	0.0009	0.0084	0.0584	0.0090	0.0025
MD	0.0008	0.0002	0.7968	0.0005	0.0008	0.1698	0.0041
VB	0.0322	0.0005	0.0050	0.0837	0.0615	0.0514	0.2231
JJ	0.0366	0.0004	0.0001	0.0733	0.4509	0.0036	0.0036
NN	0.0096	0.0176	0.0014	0.0086	0.1216	0.0177	0.0068
RB	0.0068	0.0102	0.1011	0.1012	0.0120	0.0728	0.0479
DT	0.1147	0.0021	0.0002	0.2157	0.4744	0.0102	0.0017

	Janet	will	back	the	bill
NNP	0.000032	0	0	0.000048	0
MD	0	0.308431	0	0	0
VB	0	0.000028	0.000672	0	0.000028
JJ	0	0	0.000340	0	0
NN	0	0.000200	0.000223	0	0.002337
RB	0	0	0.010446	0	0
DT	0	0	0	0.506099	0